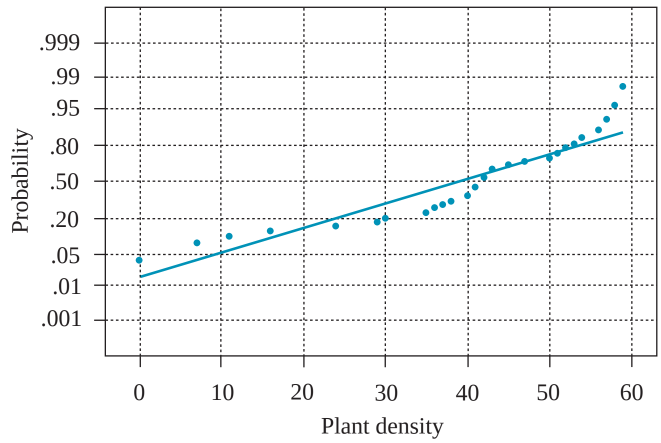
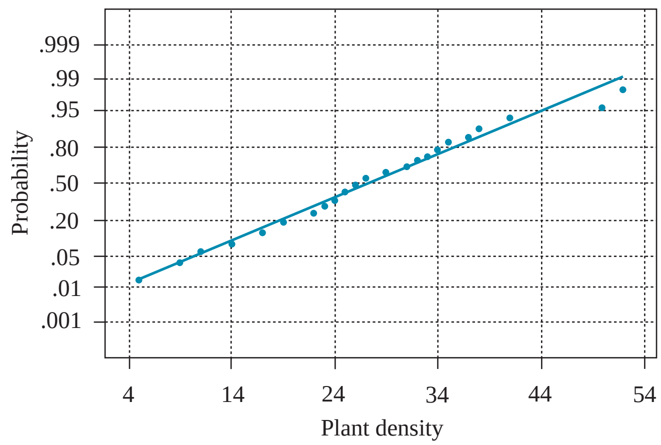
FIGURE 6.6
(a) Normal probability plot for control sites. (b) Normal probability plot for oil spill sites



(a)



(b)

t procedures will be the most appropriate inference procedures. The sample data yielded the following values:

Control Plots	Oil Spill Plots
$n_1 = 40$	$n_2 = 40$
$\bar{y}_1 = 38.48$	$\bar{y}_2 = 26.93$
$s_1 = 16.37$	$s_2 = 9.88$

The research hypothesis is that the mean plant density for the control plots exceeds that for the oil spill plots. Thus, our approximate t test is set up as follows:

$$H_0: \mu_1 \leq \mu_2 \text{ versus } H_a: \mu_1 > \mu_2$$

That is,

$$H_0: \mu_1 - \mu_2 \leq 0$$

$$H_a: \mu_1 - \mu_2 > 0$$

$$\text{T.S.: } t' = \frac{(\bar{y}_1 - \bar{y}_2) - D_0}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}} = \frac{(38.48 - 26.93) - 0}{\sqrt{\dfrac{(16.37)^2}{40} + \dfrac{(9.88)^2}{40}}} = 3.82$$